Youness HAMDI

TECHNICAL LEAD – COMPUTER VISION & AI

I solve whatever problem I am given, and I move the team forward.

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Summary

Engineer with 7 years in computer vision and AI, from deep learning to C++ optimisation, embedded cameras, robotics and measurement geometry. Whatever the problem, I start by understanding it and finding the real cause, then I decide on evidence, including the decision not to ship. I listen to the people who live with the problem, review my colleagues' work and build tools the team adopts. I now want to lead: own the technical decisions and help a team move faster.

Selected problems solved

- **A module saturating the CPU**: Wrong-way detection ran a tunnel server at 100% CPU. I replaced brute-force matching with FLANN (LSH), retuned it over 330 runs, and raised F1 from 0.45 to 0.85–0.92 on 16 sites. Shipped in several releases.
- **A detector that did not exist**: Cyclist and urban-class detection: built the dataset, then retrained SSD Lite MobileNet v2 (F1 0.80 on 6 classes, 0.72 on unseen sites). My recipe was reused for the final model.
- **A feature that should not ship yet**: Built counting zones that place themselves. Measured them on 1,730 vehicles, found the losses came from the tracker upstream, and recommended not shipping.
- **Replacing loops buried in the road**: Presence detection at intersections from video, five use cases, from scratch to release: 88.5% of events detected with 0.8 s mean latency.

How I move a team forward

- **Decide on evidence**: Every change comes with a proof run before and after it; I say clearly what is observed and what is only supposed.
- **Raise the standard without imposing it**: Test harnesses, visual test proofs (TestViz), a sequence explorer: teammates adopted them on their own.
- **Support others**: Code reviews and validation of colleagues' fixes; results and methods explained to teams and clients.
- **Organise the work**: Designed a six-phase engineering workflow run with an AI agent (Claude Code), with its agents, skills and safety hooks.

Experience

- **Computer Vision Engineer** — Bagneux, France
  Citilog - Full-time — *Sep. 2023 - Present*
  - Wrote and shipped four production analytics modules (C#/.NET): traffic statistics (rewritten from C++), city traffic with 8 classes, presence detection at intersections, and automatic road mapping for PTZ cameras.
  - Defined the acceptance thresholds of the autotests after the NF P 99-330 standard; built the annotation and ground-truth comparison tool used by the test team.
  - Vehicle length estimation and camera-height self-calibration (within 1 m on 20 real sites).
- **Software / Vision Engineer** — Paris, France
  Forssea Robotics - Full-time — *Dec. 2022 - Aug. 2023*
  - Marker-based perception and camera pose for underwater systems (ROS2, OpenCV, C++); vision integration and CI/CD.
- **Image Processing Engineer** — Bagneux, France
  Citilog - Full-time — *Dec. 2019 - Aug. 2021*
  - Wrong-way detection module (C++/OpenCV); Python multiprocessing test harness against ground truth; noise detection prototype (PSNR/SSIM).
  - Deep learning for urban counting: dataset tooling, SSD Lite MobileNet v2 retraining, evaluation of the Axis ARTPEC-7 embedded detector.
- **AI Engineer / Internships** — Paris – Lyon, France
  Groupe Complus, Niji, Kitware — *2017 - 2019*
  - Candidate-job matching models; retail product recognition and a chatbot; point cloud classification for VeloView.

Skills & Education

- **Stack**: C#/.NET, C++, Python, JavaScript; OpenCV, TensorFlow, PyTorch; profiling, testing at scale, Git, Azure DevOps, CI/CD, Docker, Claude Code
- **Epitech Paris, 2018**: Master's Degree in Information Technology. Coursera: Machine Learning (Stanford), Deep Learning (DeepLearning.AI)
- **Languages**: French (native), English (C1), Arabic / Darija (fluent), Japanese (JLPT N4)
- **Interests**: Orchestral violin; Kung-Fu / Sanda since 2012 (competition); La Bagarre, combat sports media